





HEALTHY SKIN TRIAGE BOT



**A TRIAGE SYSTEM,
NOT A DIAGNOSTIC
SYSTEM**



COMPUTER VISION

Risk detection

+

KNOWLEDGE RETRIEVAL

Clinical guidance





CONVOLUTIONAL NEURAL NETWORKS + RETRIEVAL AUGMENTED GENERATION

CNN

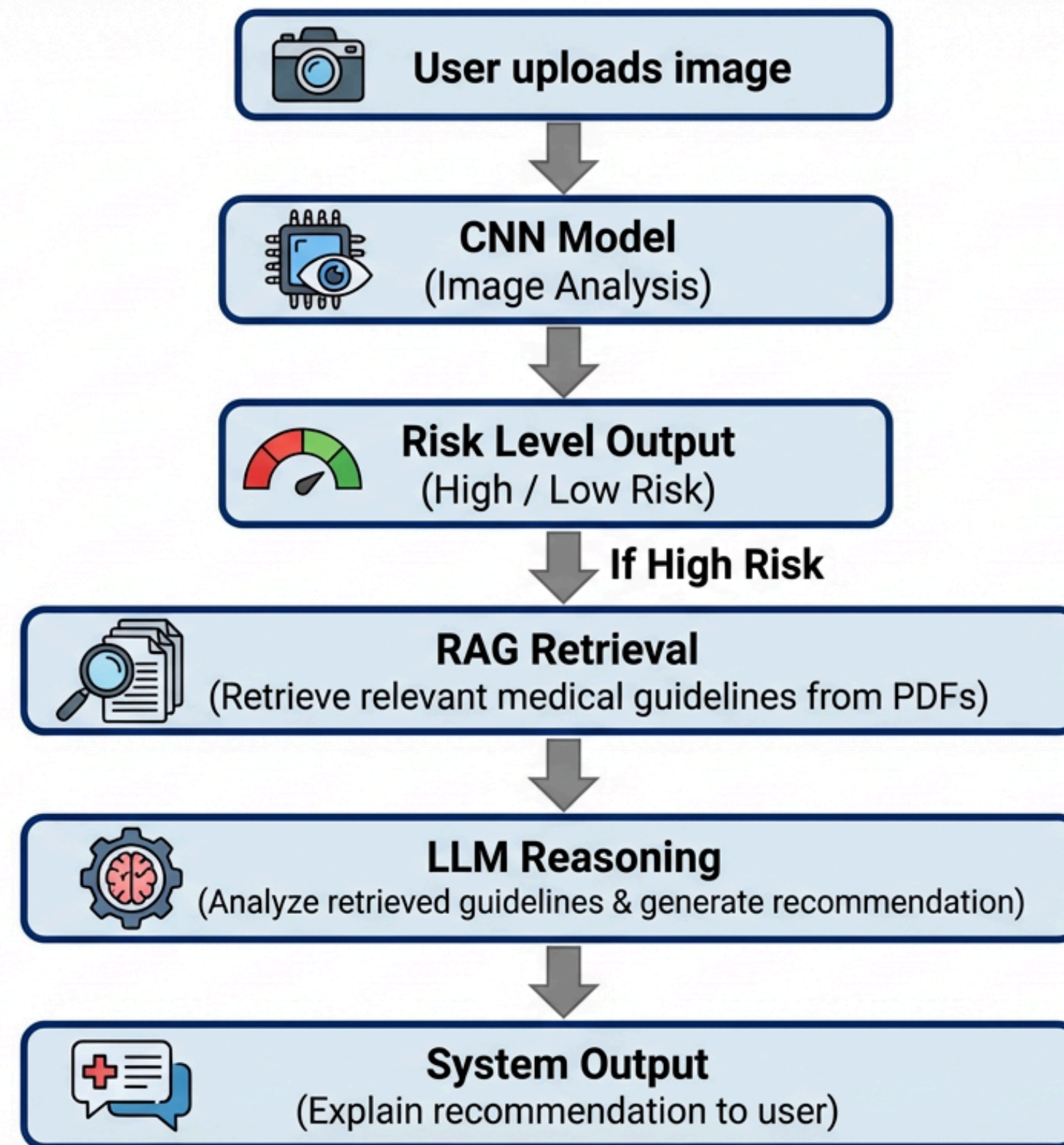
Risk detection

RAG

Clinical guidance



END TO END AI SYSTEM





INTRODUCED GUIDED LLM





WHY?





PERCEPTION + KNOWLEDGE RETRIVAL

CNN

RAG

=

RAW RESULTS





PERCEPTION + KNOWLEDGE RETRIVAL + REASONING

CNN

RAG

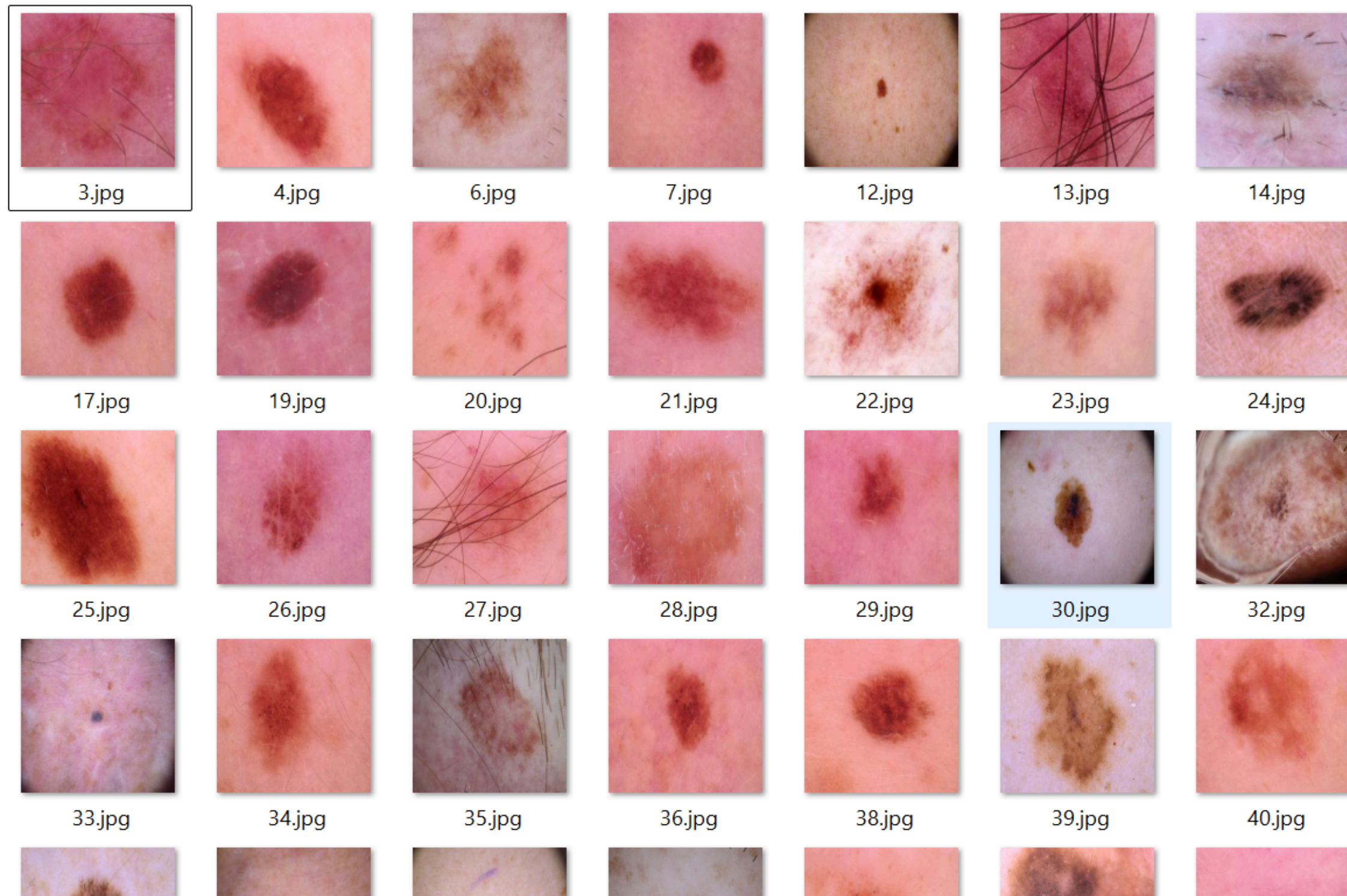
LLM

=

HUMAN READABLE RESULTS



DATASET

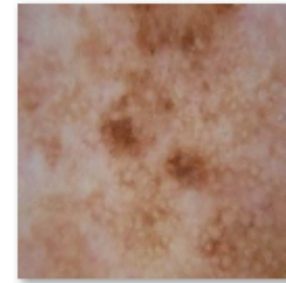


benign (1,440 items)

i Select a single file to get more information and share your cloud content.



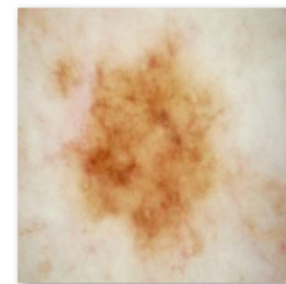
A decorative graphic in the top right corner of the page. It features a large, dark gray, semi-circular shape on the right, and a series of concentric, dark gray circular lines on the left, resembling a stylized eye or a target. The lines are of varying thickness and are arranged in a way that suggests depth and movement.



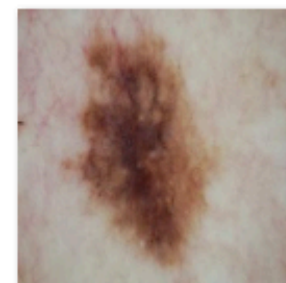
11.jpg



20.jpg



28.jpg

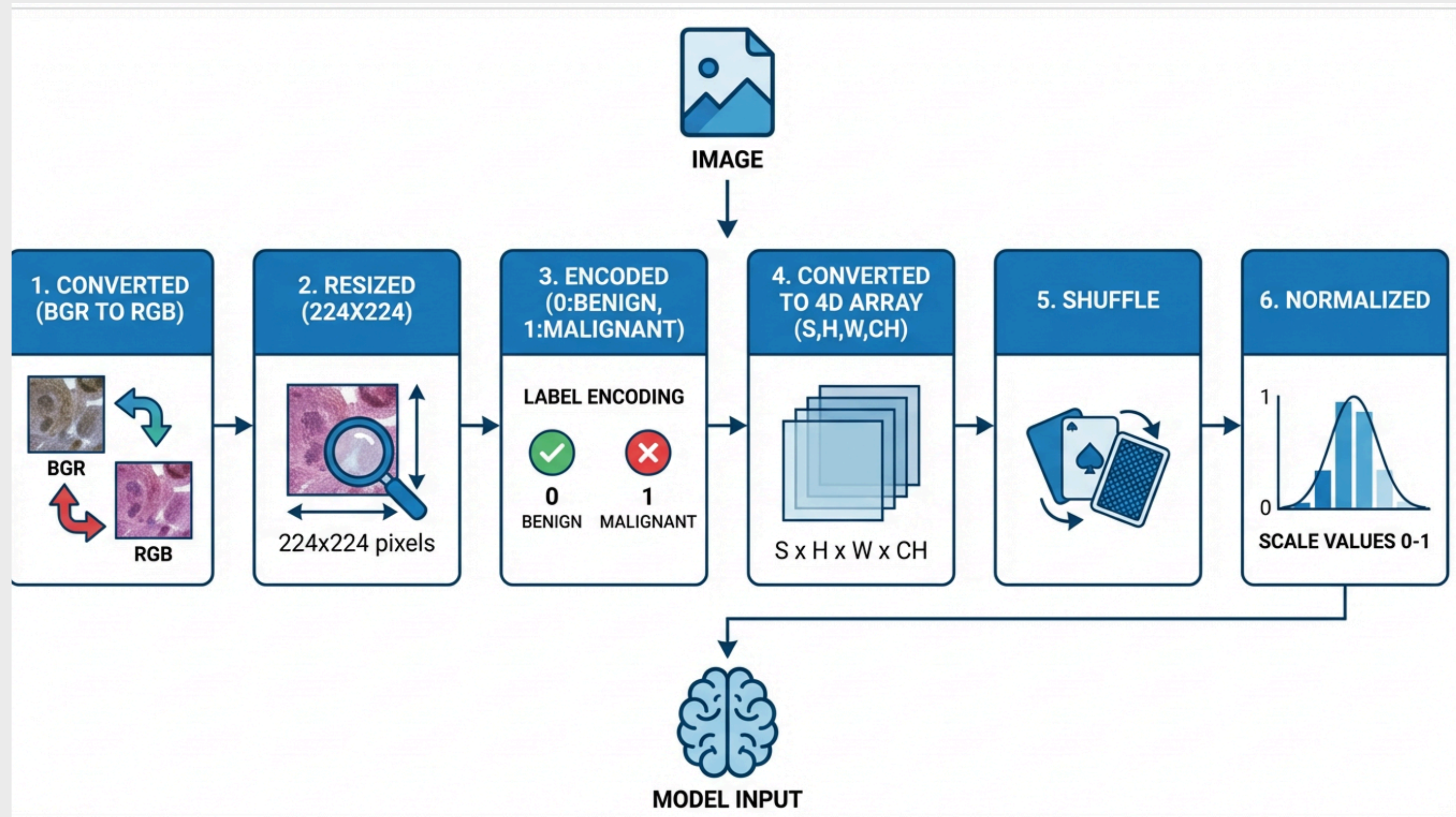


36.jpg



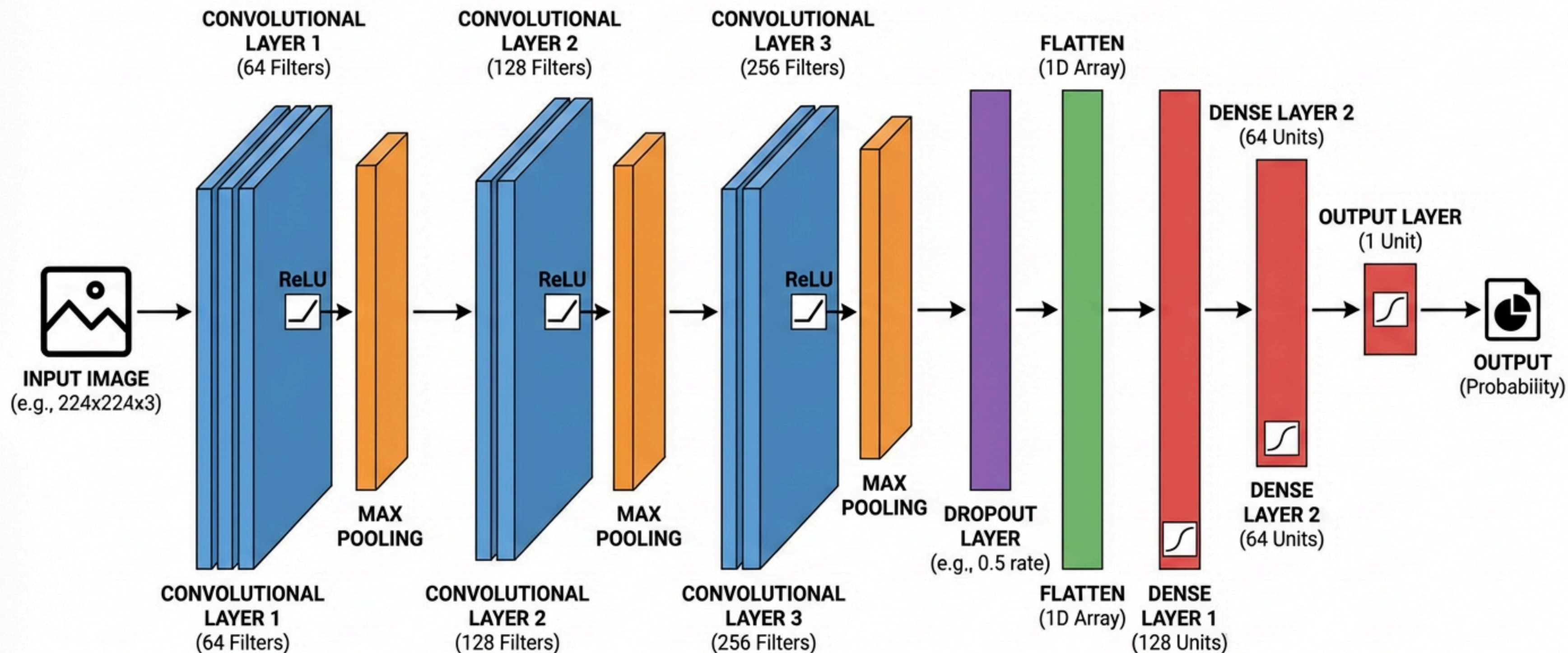
Select a single file to get more information and share your cloud content.

COMPONENT A: CNN



MODEL ARCHITECTURE

CNN MODEL ARCHITECTURE



EARLY STOPPING MECHANISM

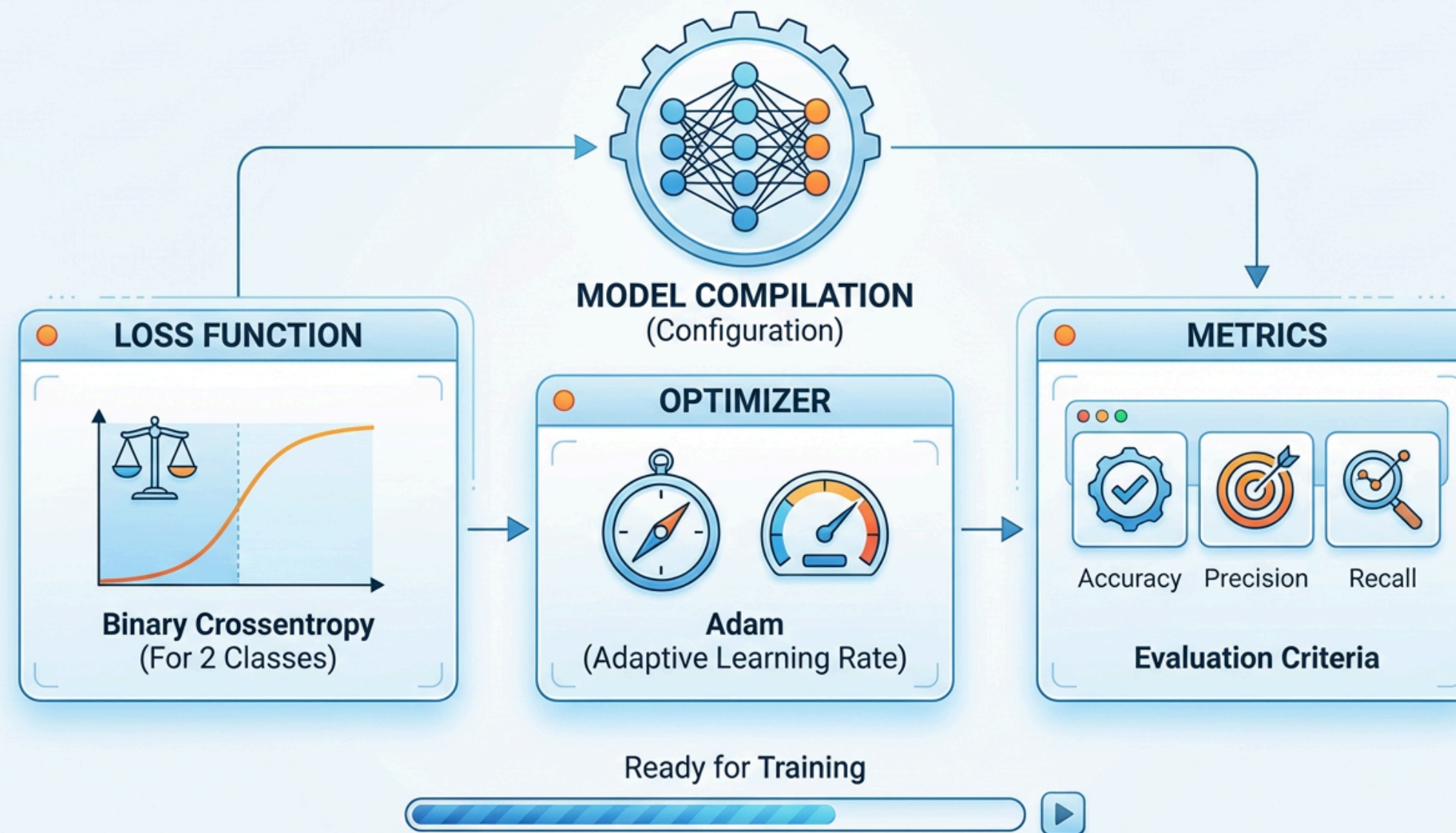


EarlyStopping: Prevent Overfitting

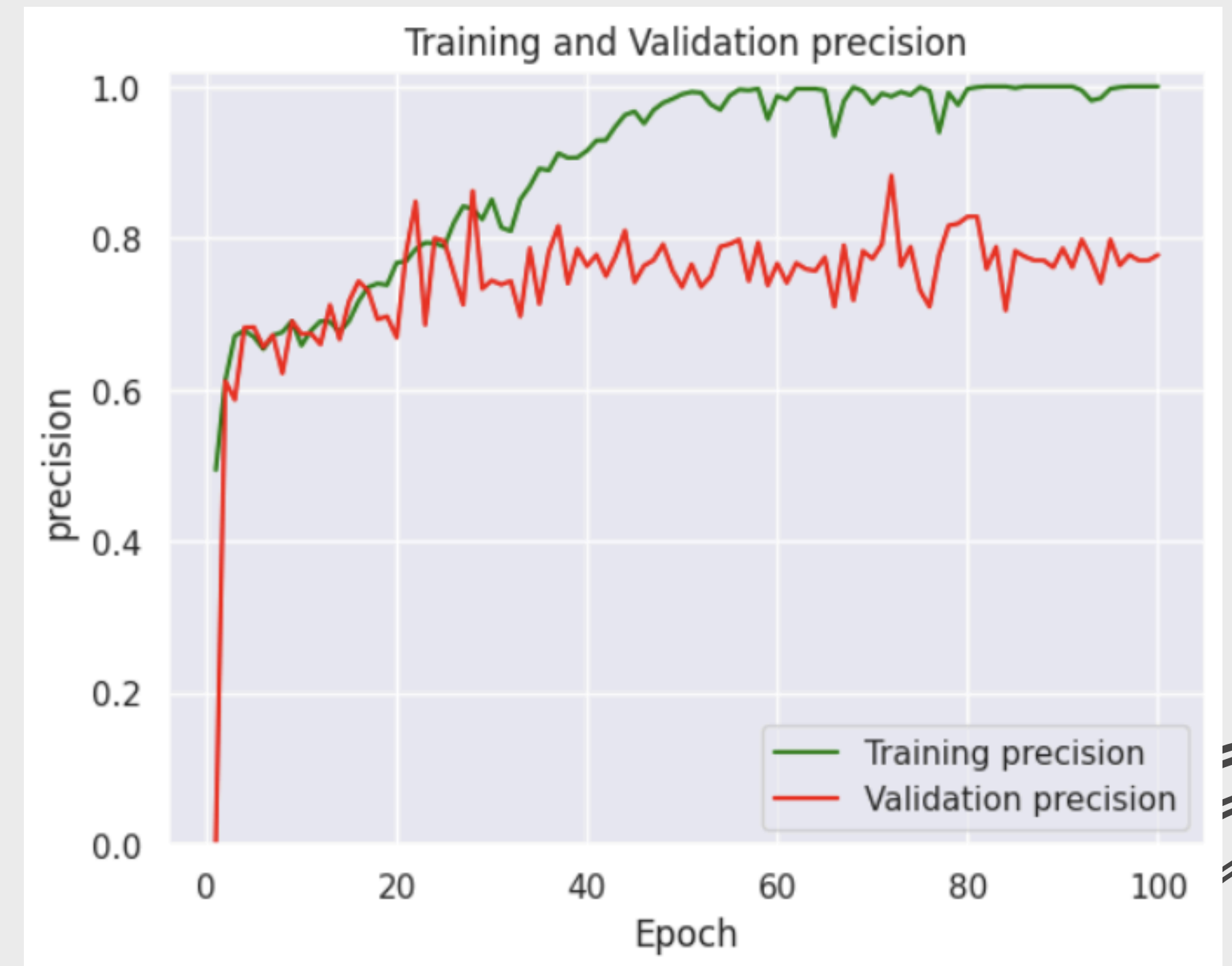
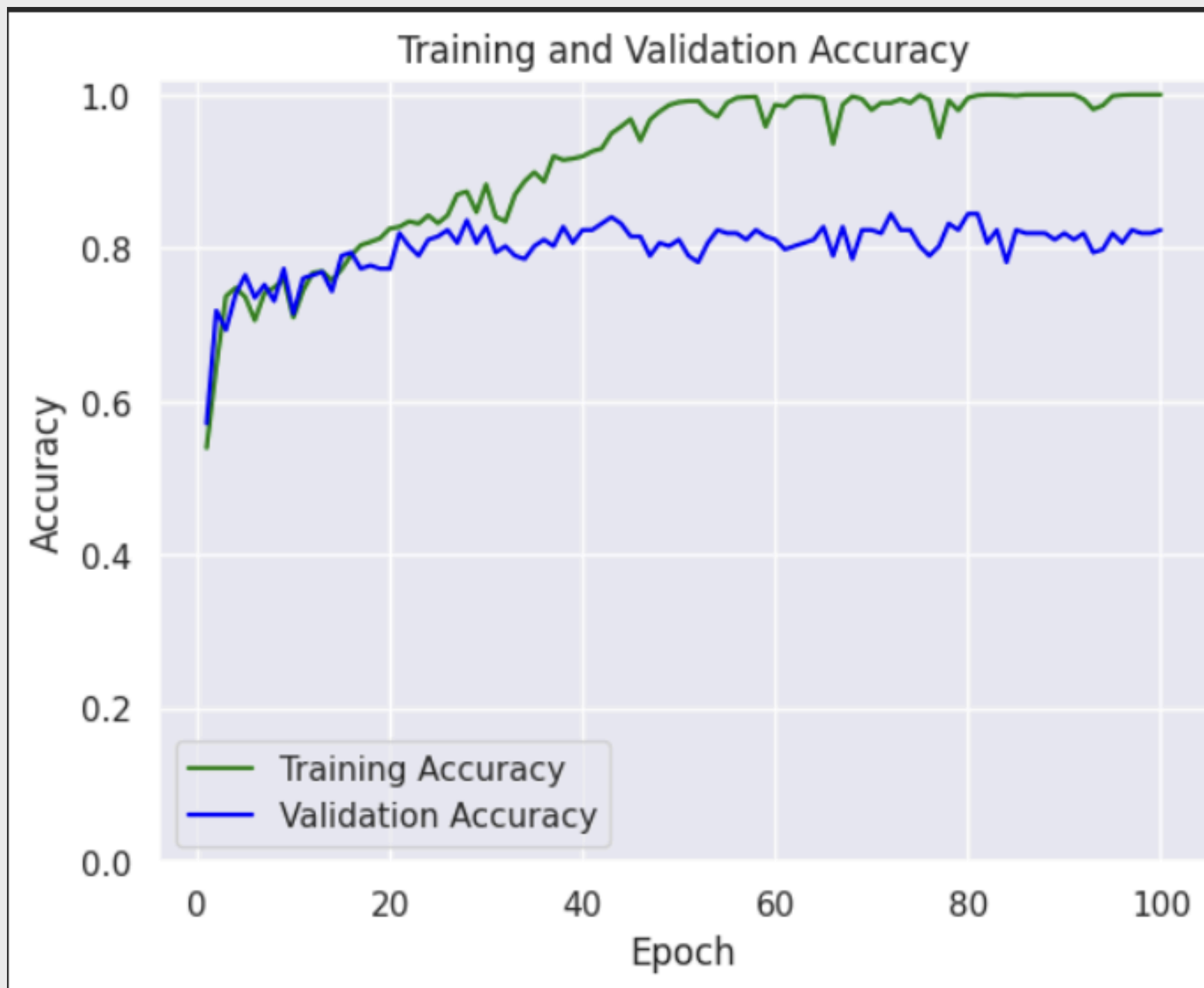
Monitors a metric (e.g., `val_loss`).
Stops training when metric worsens for a
“patience” period, preventing model
from learning noise.

COMPILATION

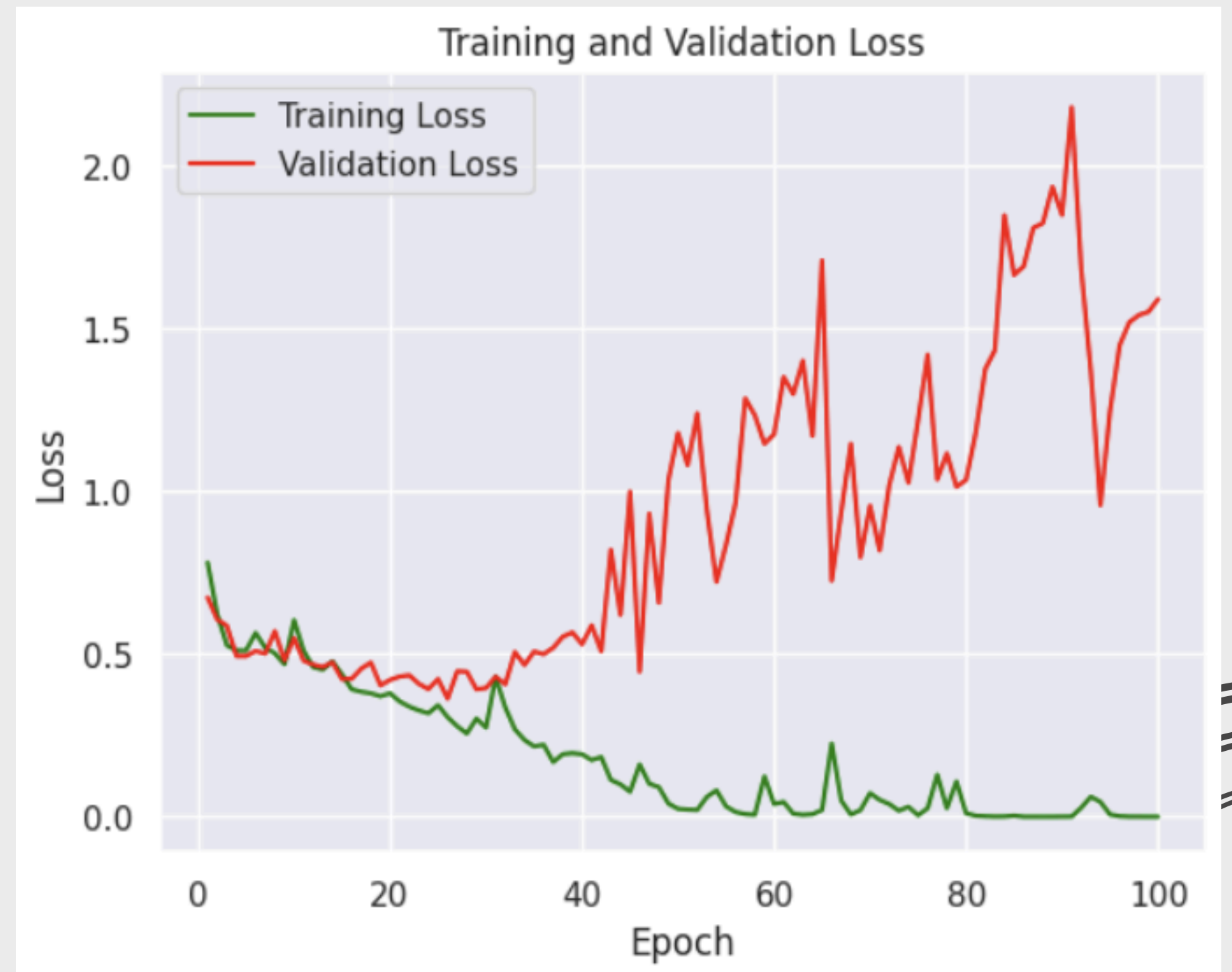
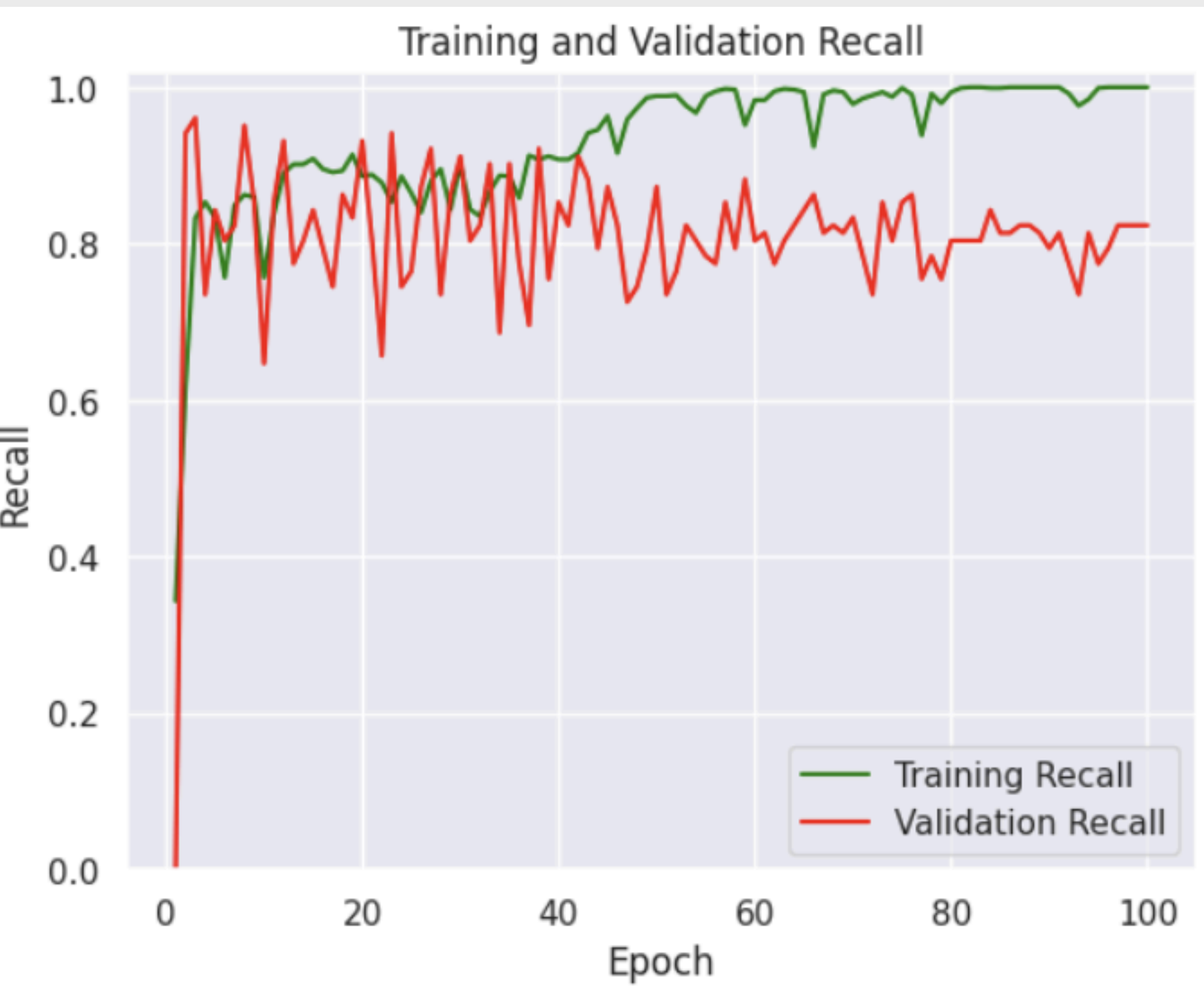
```
# Compiling the model  
model.compile(loss="binary_crossentropy", optimizer= 'Adam', metrics=["accuracy", "precision", "recall"])
```



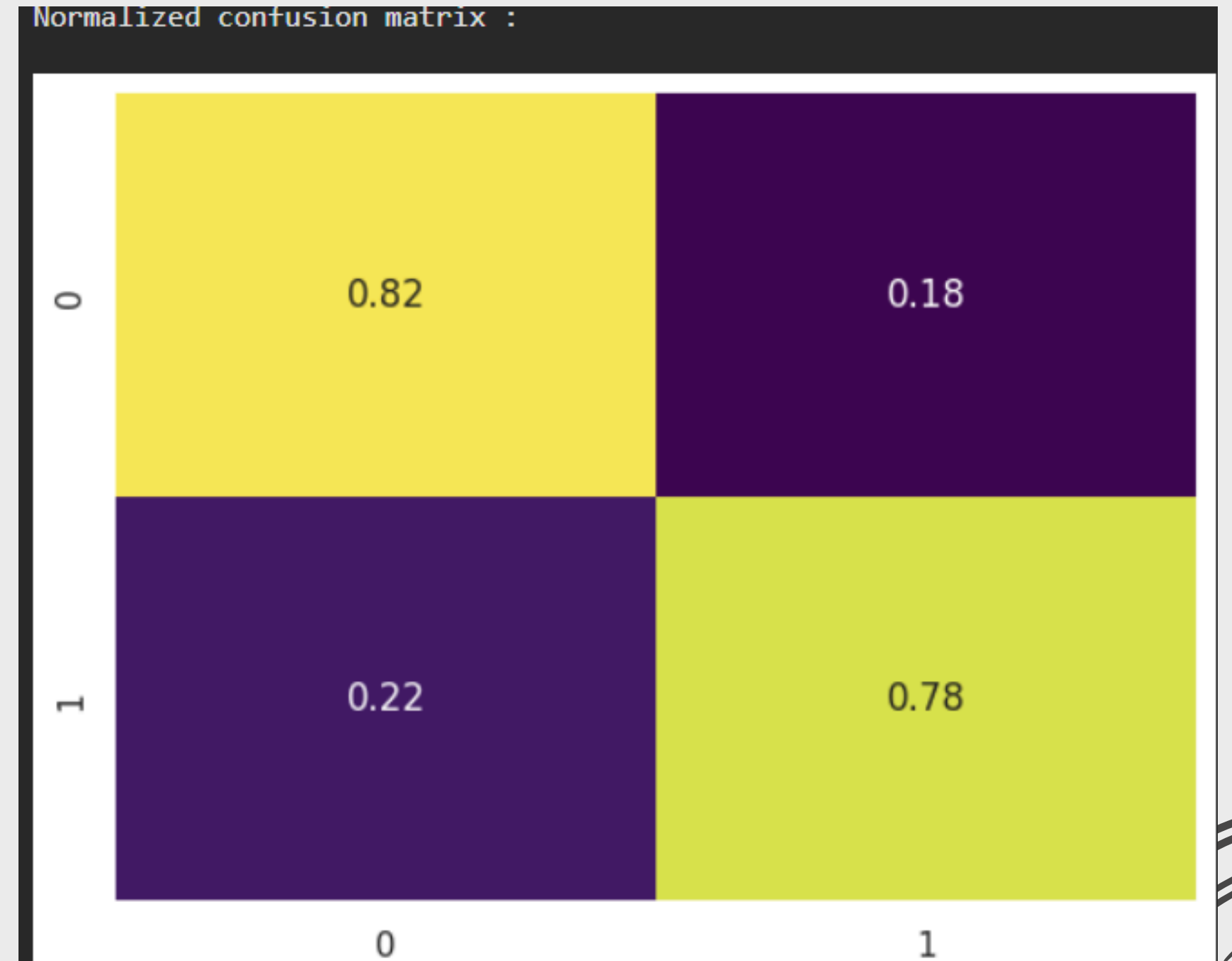
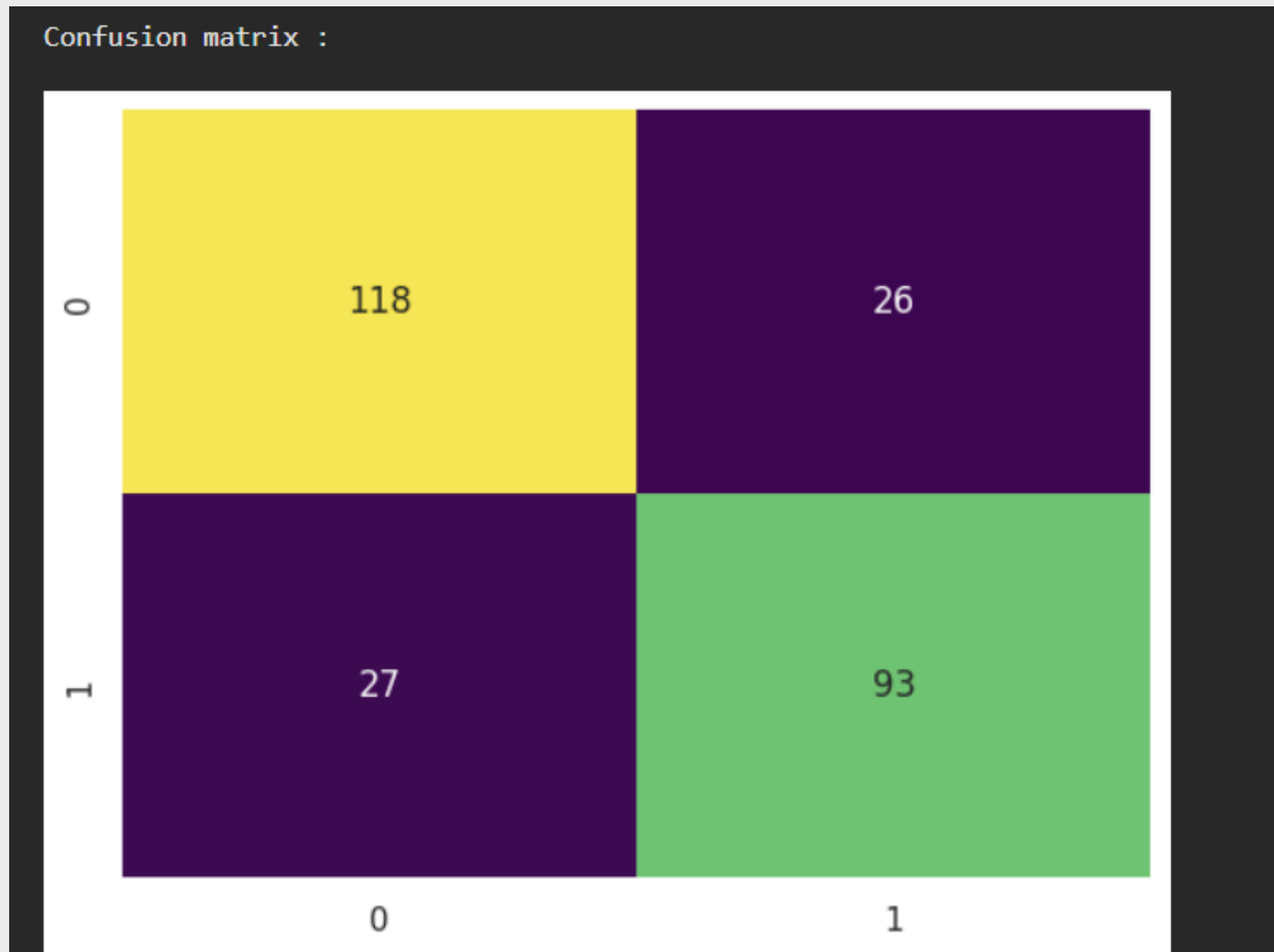
EVALUATION METRICS



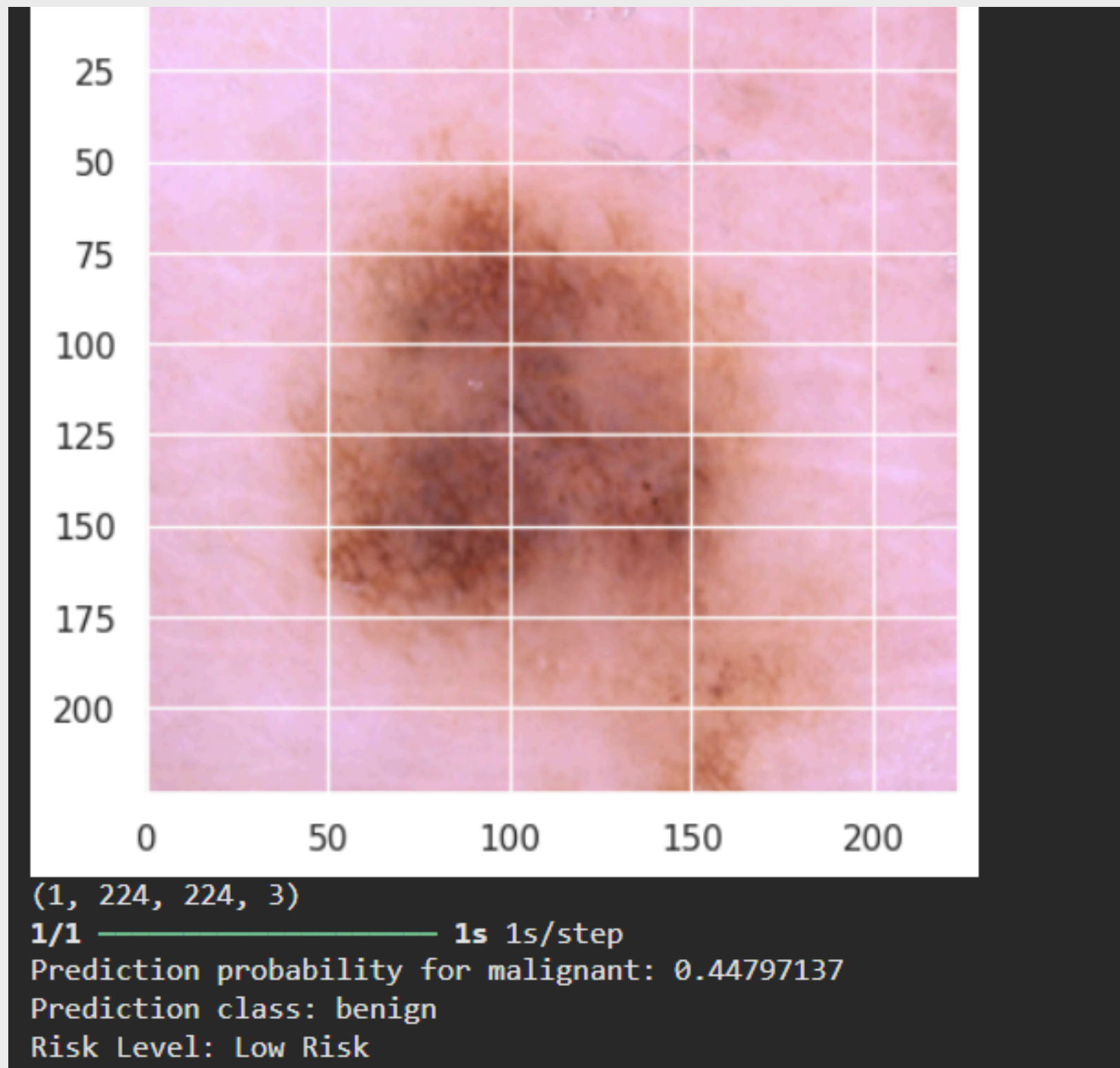
EVALUATION METRICS



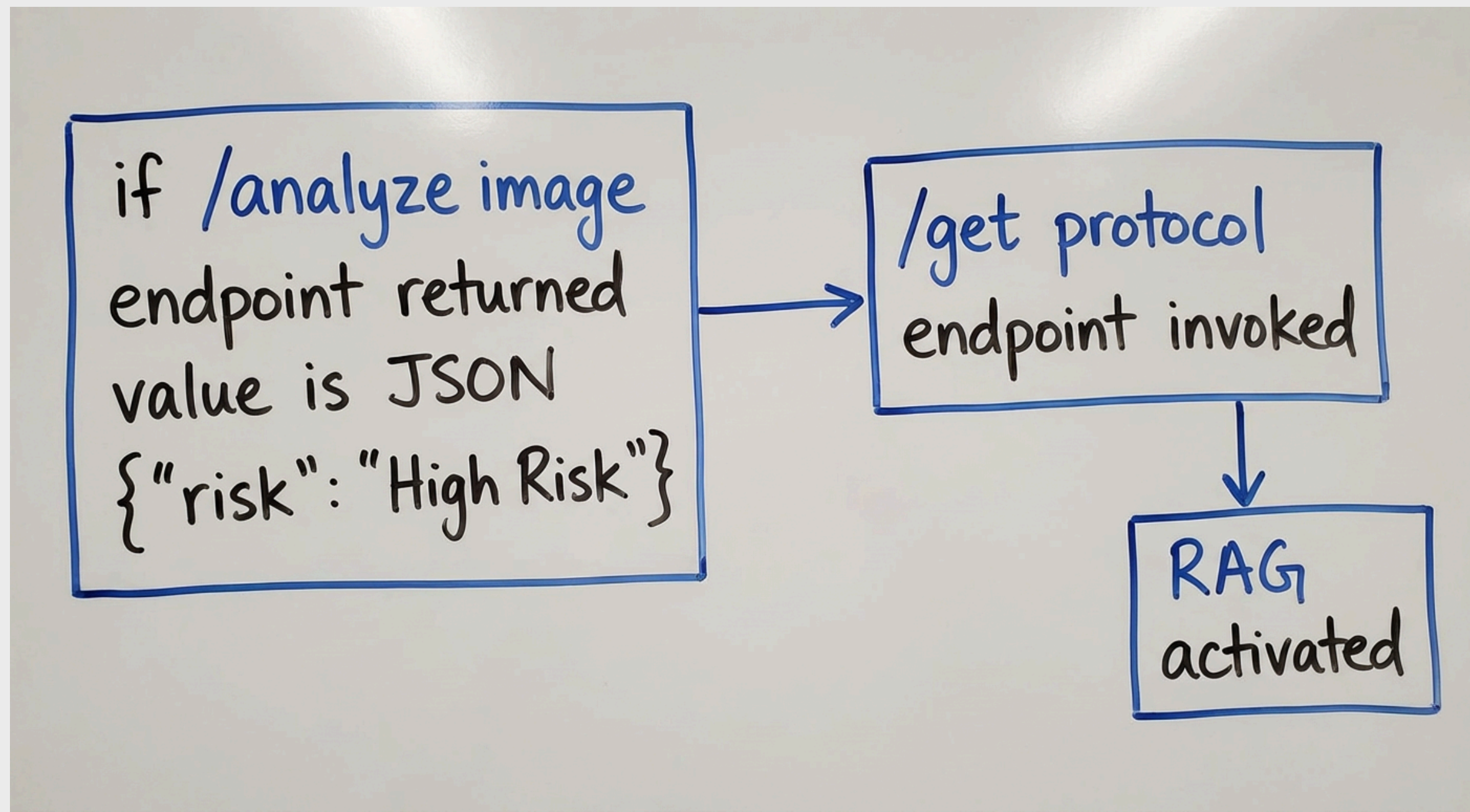
CONFUSION MATRIX



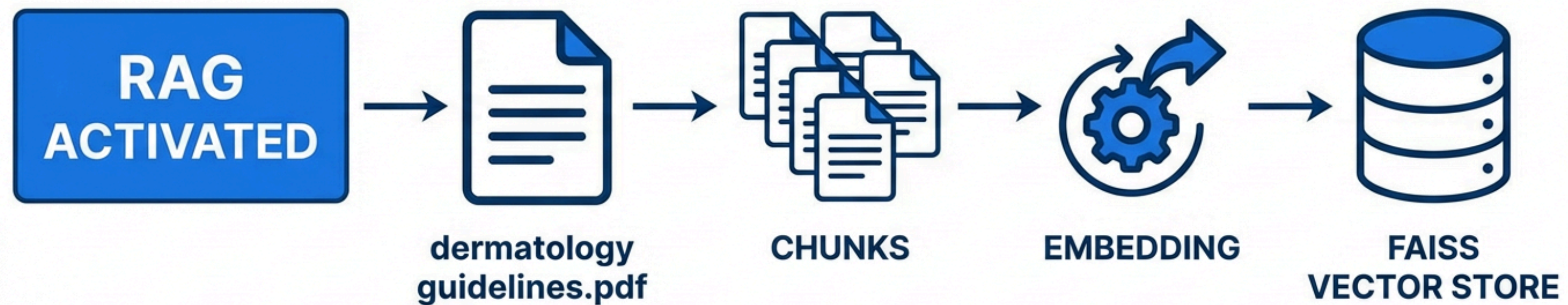
PREDICTION SAMPLE



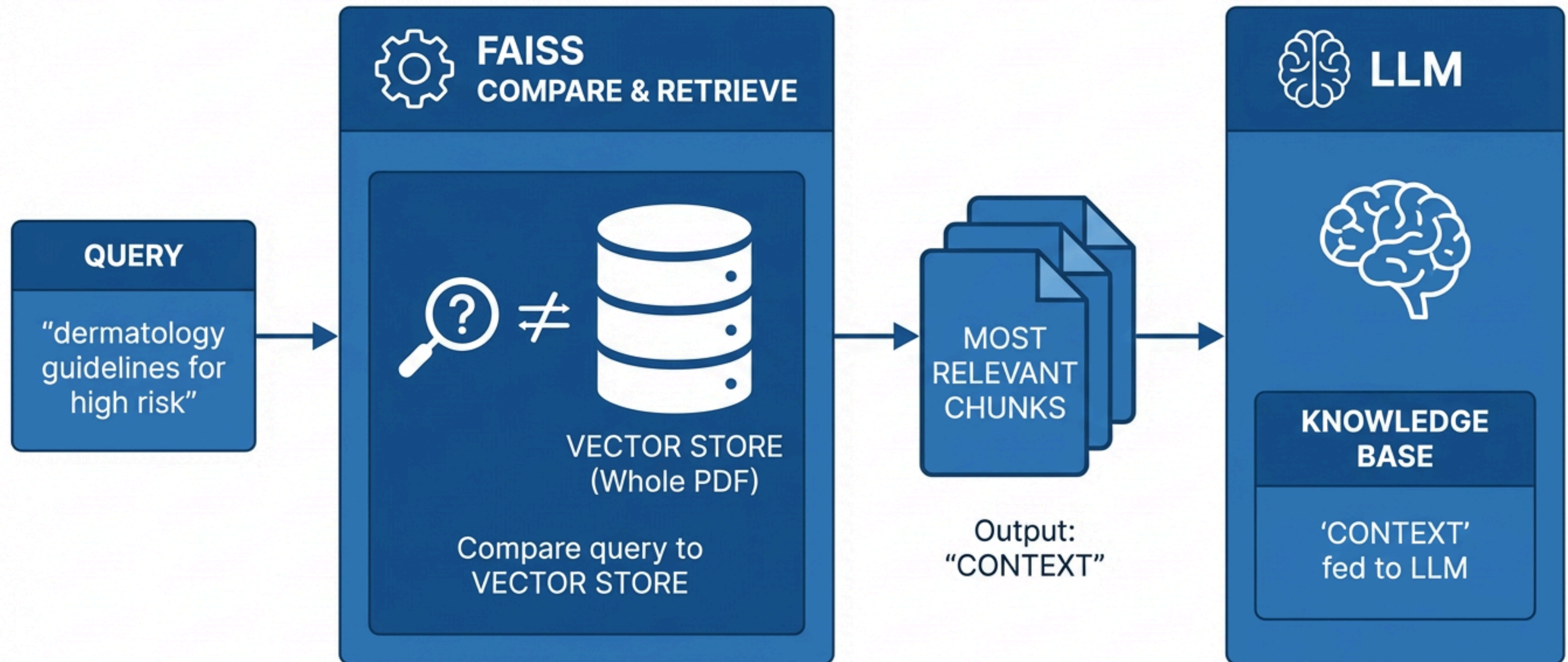
CONDITIONAL TRIGGER FOR COMPONENT B: RAG



RAG ACTIVATED



RAG ACTIVATED





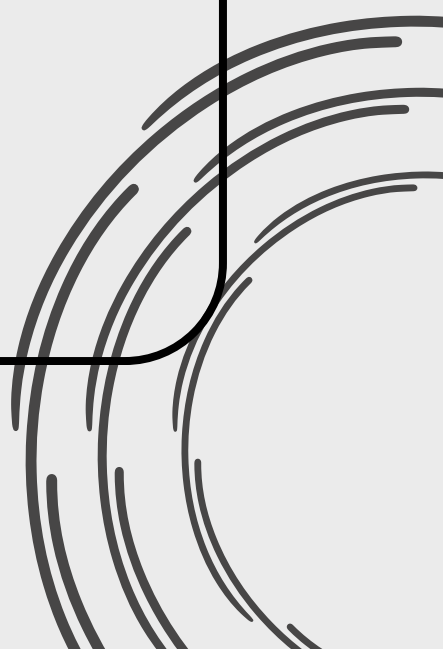
API DESIGN

1- /analyze - image > perception

2- / get - protocol > reasoning

FRONTEND

Streamlit



RESULT



Healthy Skin Triage Bot

Upload skin lesion image



Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files



53.jpg 17.7KB



Uploading image...

Status code: 200

Raw response: {"risk":"High Risk","confidence":0.5816097259521484,"explanation":"The result from the assessment suggests that the skin spot we looked at has a high chance of being a type of skin cancer called melanoma, which is serious and needs to be checked and treated by a doctor. Melanoma can grow deeper into the skin and may spread if not treated early. \n\nIt's important to see a dermatologist as soon as possible so they can examine the spot more carefully and perform tests, such as a biopsy, to find out exactly what it is. Early treatment of melanoma has a very good chance of success \n\nMedical disclaimer:

RESULT

Prediction successful!

▼ { 

"risk" : "High Risk"

"confidence" : 0.5816097259521484

"explanation" :

"The result from the assessment suggests that the skin spot we looked at has a high chance of being a type of skin cancer called melanoma, which is serious and needs to be checked and treated by a doctor. Melanoma can grow deeper into the skin and may spread if not treated early.

It's important to see a dermatologist as soon as possible so they can examine the spot more carefully and perform tests, such as a biopsy, to find out exactly what it is. Early treatment of melanoma has a very good chance of success.

Medical disclaimer: This information is for educational purposes and does not replace professional medical advice. If you have concerns about your skin or any skin changes, please consult a qualified healthcare professional promptly."





LIMITATIONS

- Dataset was limited
 - If pdf guidelines are outdated or incomplete this will affect recommendation
- 



FUTURE WORK

- Train on larger and more diverse datasets
 - Multi-Class & Severity Classification
- 



THANK YOU

